

Spherification (Direct Method)

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This is a method for spherification. Making those little, caviar-like pearls of liquid you see in high-end cuisine. There are a few main methods for spherification. Cold oil spherification will give you a product that is largely solid/gelatin throughout. Reverse spherification will produce a membrane on the exterior of the sphere. The membrane can be a bit thick and is great for making larger spheres. This method is direct spherification. Much like reverse spherification, the end product should be a sphere with an exterior membrane and a liquid center. The membrane for this method is much thinner and is a great method for smaller spheres, such as the ones pictured

Step 1: Gather Your Ingredients



For direct spherification, you only need a few ingredients. Two are specialty ingredients which can be bought easily on Amazon or from Modernist Pantry. You will need:

- Sodium Alginate
- Calcium Chloride
- A liquid of your choosing.
You can use juices, thin purees, etc. (I diced and juiced the pictured watermelon)
- (For acidic ingredients) Sodium Citrate

Step 2: Gather Your Equipment



You will need:

- A blender (I'd recommend an immersion blender unless you have a vacuum chamber)
- Two scales, one of which can measure to the tenth of a gram
- A pH meter or pH test strips (Only necessary if making spheres out of something acidic)
- Bowls
- A sieve
- A food syringe or squirt bottle

Step 3: Test and Alter the PH



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Preparing the Calcium Bath for Basic Spherification

While you wait for the main ingredient to settle in the fridge, prepare the calcium bath by dissolving the calcium salt in a bowl to obtain a solution with 0.18% calcium. Keep in mind that different calcium salts contain different amounts of calcium ions and therefore the amount you use to create the calcium bath will be different too. The most common calcium salts used in spherification and the concentrations to create the calcium bath are:

Calcium Salt	Calcium Content	Qty to Make 0.18% Ca solution
Calcium Chloride	36.1%	0.5%
Calcium Lactate	18.4%	1%
Calcium Lactate Gluconate	9.3%	2%

-0.5% Calcium Chloride: this is the preferred calcium salt to create the bath for Basic Spherification. Use 0.5 g per 100 g of water to create a 0.5% solution of Calcium Chloride. You can just stir it since the Calcium Chloride dissolves very easily in water. Calcium Chloride has a bad salty flavor but you can rinse it off with water. Calcium Chloride is very hygroscopic (absorbs water in the air quickly) so close the container quickly, store it in a dry place and consider using a [desiccant packet](#) if the humidity in the air is high. Calcium Chloride is included in the [Molecular Gastronomy Essentials Kit](#), the [Molecular Gastronomy Premium Kit](#) and the [Molecular Gastronomy Ultimate Kit](#).

-1% Calcium Lactate: mostly used for Reverse Spherification because it has better flavor. Use 1 g per 100 g of water to create a 1% solution of Calcium Lactate. You can stir it or mix with a blender.

-2% Calcium Lactate Gluconate: the preferred calcium salt for Reverse Spherification because it has no discernible flavor. Use 2 g per 100 g of water to create a 2% solution of Calcium Lactate. You can stir it or mix with a blender. Calcium Lactate Gluconate is included in the [Molecular Gastronomy Essentials Kit](#), the [Molecular Gastronomy Premium Kit](#) and the [Molecular Gastronomy Ultimate Kit](#).

Increasing the Calcium Bath Density

To aid the formation of spherical shapes when you pour the alginate solution into the bath, you can increase the density of the bath to prevent the spheres from flattening at the bottom of the container. You can achieve this by adding sugar to the calcium bath up to 20%.

Creating the Spheres in the Calcium bath

If you are using something acidic, you're going to want to test the pH. If you aren't, this step isn't 100% necessary. If the pH is below 3.6, the chemical reaction that makes direct spherification possible won't happen the way you want it to. If you are only slightly below 3.6, you can add sodium citrate to increase the pH. Keep in mind, sodium citrate does have a taste. If you add too much, you will end up with an undesirable final product. If you can't bring the pH high enough for direct spherification, I'd recommend using reverse spherification for a similar product where pH is less important.

Step 4: Weigh Out Your Ingredients



Direct spherification, much like many other techniques in molecular gastronomy, requires precision for the desired outcome. This is why we're weighing things out.

1. Weigh out whatever liquid you are making spheres out of.
2. Weigh out 0.5% of your liquid's weight in sodium alginate
3. Weigh out the amount of water you are going to use
4. Weigh out 1% of the weight of your water in calcium chloride.

If you have 1000 grams of juice, you'd want 5g of sodium alginate.

If you have 1000 grams (1L) of water, you'd want 10g of calcium chloride.

Step 5: Blend Your Sodium Alginate Into the Liquid



After you've got everything weighed out, you can start putting things together. If you don't have a vacuum chamber, I highly

recommend using an immersion blender for this step to incorporate less air and reduce the resting time at the end of it.

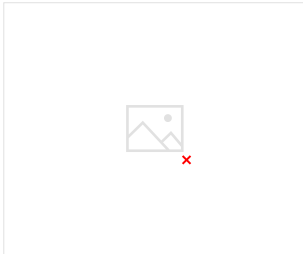
If making spheres out of a thin liquid like a juice, I'd recommend adding the sodium alginate to a small portion of a third or a half of the liquid and blending it together, then adding that to the remaining liquid. When blending, blend for a couple minutes to ensure the sodium alginate is well mixed with the liquid. This is going to help disperse the sodium alginate in the liquid.

If you are making spheres using alcohol, blend the sodium alginate into some distilled water and then add that water to the alcohol. Much like with the thin liquids mentioned above, this will assist in the dispersion of the alginate.

If you are using a thicker liquid, feel free to just go ahead add the sodium alginate to the full amount and blend.

Once everything is blended, let it sit for a few minutes to hydrate and then pour it through a sieve. This will ensure you have no gummy strands clogging things up later in the process.

Step 6: Let It Rest



At this point, you're going to have a somewhat aerated liquid. You don't want that air. Air will make the liquid float, preventing full immersion in the calcium bath you're about to make and also make your spheres misshapen. Take this liquid and throw it in the fridge for an hour or two to allow the air to make its way out.

If you have a vacuum chamber, you don't need to wait and you can enjoy the next step.

Step 7: Remove Air Using a Vacuum Chamber (if You Have One)

For this step, put your liquid in a high walled container and use your vacuum chamber to draw the air right out of the liquid. This is my favorite step. You will understand if you watch the video.

Step 8: Make Your Calcium Bath and Prepare a Clean Water Bath



Add your calcium chloride into the water you are going to use and mix until dissolved. In another bowl, pour in some clean

water. We're going to use this to clean off the calcium chloride from the outside of the spheres. I'd recommend having two clean water baths. One to rinse off the spheres and one to hold them in afterward. The second clean water bath will also be cleaner than the first and will not have any of the calcium chloride flavor

Step 9: Make Your Spheres



Once your alginate mixture and calcium bath are prepared and are air free, you're ready to make your spheres. Fill up your syringe or squeeze bottle with your flavored liquid, hold it above the calcium bath, and then slowly squeeze the liquid in. I'd recommend doing this about 3-6 inches above the water bath. This will help create a more perfect sphere. I'd also recommend moving the bottle around in a spiral above the water bath while doing this to ensure you don't end up with spheres sticking together. The sodium alginate and calcium will react, forming a thin membrane on the outside of the sphere.

The reaction will take about a minute or so to get to the point where these spheres will have enough structural integrity to be used. The longer you leave them in the bath, the thicker the membrane of the spheres will become. While they are in the calcium bath, gently stir them occasionally. A spoon with holes in it is best for this, but not necessary.

After approximately a minute, use either a strainer or slotted spoon (they sell spoons specifically for spherification. I'd recommend one) to transfer your spheres from the calcium bath to your clean water bath. Stir them around a bit here to help remove calcium chloride from the exterior. Once rinsed, transfer the spheres to a new clean water bath for holding and use as desired.